

the heart of quantum interference and many of the remarkable advantages offered by quantum computing. Now, let us execute three related Qiskit programs to explore the effect of the Pauli-Z gate.

The first program, *'qiskit3.py'*, was introduced in the previous article. It applies a Hadamard gate to a qubit and then measures the qubit to demonstrate how the Hadamard gate creates an equal superposition of the $|0\rangle$ and $|1\rangle$ states.

Let us now examine a quantum program that demonstrates the effect of the Pauli-Z gate on a qubit. Consider the program *'qiskit5.py'*, shown below. Starting from the initial state $|0\rangle$, the program first applies a Hadamard gate to create an equal superposition of $|0\rangle$ and $|1\rangle$. It then applies a Pauli-Z gate, followed by a measurement of the qubit. By comparing the output of this program with that of the previous one, we can investigate whether the Pauli-Z gate has any observable effect on the measurement outcome. Notice that the only new line of code requiring explanation is *'qc.z(0)'*, which applies a Pauli-Z gate to the single qubit in the circuit.

```
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator
```

```
qc = QuantumCircuit(1, 1)
qc.h(0)
qc.z(0)
qc.measure(0, 0)
```

```
sim = AerSimulator()

result = sim.run(qc, shots=1000).
result()
counts = result.get_counts()
print(counts)
```

The program *'qiskit6.py'*, shown below, extends *'qiskit5.py'* by inserting a second Hadamard gate after the Pauli-Z gate and before measurement. We add this second Hadamard gate because it converts the otherwise hidden phase information into measurable

probabilities, making the effect of the Pauli-Z gate observable. As we shall see, this seemingly minor modification dramatically changes the outcome of the computation.

```
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from qiskit_aer import AerSimulator
```

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qc = QuantumCircuit(1, 1)
qc.h(0)
qc.z(0)
qc.h(0)
qc.measure(0, 0)
```

```
sim = AerSimulator()

result = sim.run(qc, shots=1000).
result()
counts = result.get_counts()
print(counts)
```

Execute the three programs after activating the virtual environment *qiskit_env*, as described in the previous articles. The measurement results obtained from these programs are shown in Figure 1. What conclusions can we draw from these outputs?

```
(qiskit_env) # python qiskit3.py
{'1': 489, '0': 511}
(qiskit_env) # python qiskit5.py
{'1': 473, '0': 527}
(qiskit_env) # python qiskit6.py
{'1': 1000}
```

Figure 1: Measurement results of the three Qiskit programs

Observe that the outputs of *'qiskit3.py'* and *'qiskit5.py'* are nearly identical, with approximately equal numbers of 0s and 1s. This indicates that inserting a Pauli-Z gate before measurement does not alter the measurement probabilities. However, the output of *'qiskit6.py'* is strikingly different. After applying the second Hadamard gate, the qubit is measured exclusively in the state $|1\rangle$. Clearly, the Pauli-Z gate was not “doing nothing” after all—it was modifying the phase of the quantum state, an effect that becomes observable only

after a subsequent quantum operation. This simple experiment illustrates the essence of quantum interference: although the phase introduced by the Pauli-Z gate is invisible by itself, it changes how later quantum operations combine probability amplitudes.

To verify this behaviour visually, let us return to the Bloch sphere simulator. Perform the same sequence of operations—Hadamard, Pauli-Z, and Hadamard (H–Z–H)—on the Bloch sphere. The final configuration is shown in Figure 2. Notice that the qubit ultimately points towards the south pole, corresponding to the state $|1\rangle$, perfectly matching the measurement results produced by the program *'qiskit6.py'*.

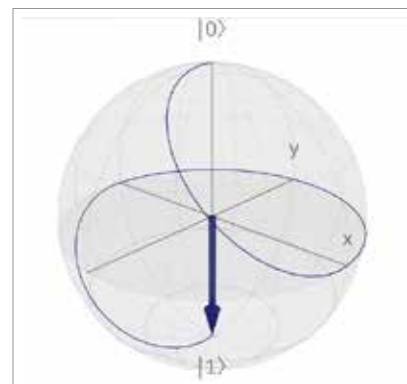


Figure 2: Evolution of the qubit state on the Bloch sphere during the H–Z–H gate sequence

Let us now turn our attention to the mathematical representation of the Pauli-Y and Pauli-Z gates. As shown in Figure 3, both gates, like the Pauli-X gate, are represented by 2×2 matrices. Using the rules of matrix multiplication, verify that multiplying the Pauli-Z matrix by the column vector representing the state $|0\rangle$ leaves the state unchanged, while multiplying it by the column vector representing $|1\rangle$ introduces a minus sign, producing the state $-|1\rangle$. Similarly, verify that multiplying the Pauli-Y matrix by the column vector representing $|0\rangle$ yields $i|1\rangle$, whereas multiplying it by the column vector representing $|1\rangle$ gives $-i|1\rangle$ (Here, i denotes the imaginary unit, defined by the property $i^2 = -1$). These